Persona Directions Reduce Forced-Choice Sycophancy via Activation Steering

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Abstract

Large language models often exhibit sycophancy, adapting responses to match a user's stated view even when it is false. We test whether general-purpose persona directions, extracted without sycophancy labels, reduce forced-choice sycophancy as effectively as a targeted Contrastive Activation Addition (CAA) baseline. On Gemma 2 27B and Qwen 3 32B, evaluated on a held-out PhilPapers A/B benchmark with Holm-corrected significance testing, critical-role vectors (Skeptic, Devil's Advocate, Judge) reach 68% (Gemma) and 116% (Qwen) of CAA's Δlogit and are Holm-significant on all test seeds; conformist roles show weak, heterogeneous effects. All role–CAA cosines are below 0.17, but the sign of $\cos(\text{role}, \text{CAA})$ flips across models, so geometric independence does not establish mechanistic independence. Critical-role steering is a lightweight, data-efficient intervention, while the polarity asymmetry complicates simple family-level predictions.

1. Introduction

Sycophancy — the tendency of large language models to agree with users regardless of factual correctness — is among the most persistent failure modes of RLHF-trained systems (Perez et al., 2022; Sharma et al., 2023). Even when a model has encoded the correct answer internally, the reward model's preference for agreement can override it (Wang et al., 2025a). How can we intervene on sycophancy without expensive retraining or curated behavioral datasets?

Contrastive Activation Addition (CAA) (Rimsky et al., 2024; Turner et al., 2023) extracts a steering vector from contrastive sycophantic/honest prompt pairs and adds a scaled version during inference. While effective, CAA requires hundreds of behavior-specific pairs and must be re-curated for each new target behavior. A natural alternative is to reuse *persona* representations already learned by instruction-tuned models (Lu et al., 2026).

We ask three questions. (i) Can off-the-shelf role vectors reduce forced-choice sycophancy comparably to CAA? (ii) Do critical/conformist family labels predict steering direction? (iii) Are effective role vectors geometrically distinct from the CAA direction? We evaluate across Gemma 2 27B (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) on a counterbalanced PhilPapers benchmark with tune/test splitting and Holm correction.

Contributions. (1) Critical-role vectors reach 68–98% of CAA's Δlogit with strong cross-seed consistency, using *no* sycophancy labels. (2) Conformist roles produce weak, heterogeneous effects, partially falsifying bidirectional family-level prediction. (3) All role vectors are nearly orthogonal to CAA ($|\cos| < 0.17$), but the *sign* of the cosine flips between Gemma and Qwen — a cross-model geometric asymmetry we make explicit as a new caveat on mechanistic-independence claims.

2. Related Work

Sycophancy in LLMs. Perez et al. (2022) introduced model-written sycophancy evaluations exposing systematic agreement bias. Sharma et al. (2023) showed sycophancy is widespread and emerges during RLHF. Wang et al. (2025a) traced sycophancy to override mechanisms in RLHF-trained models: the correct answer is often encoded internally but suppressed by preference for agreement.

Activation steering and CAA. Activation addition (Turner et al., 2023) demonstrated that fixed residual-stream vectors can steer LLM behavior at inference time. CAA (Rimsky et al., 2024) formalizes this via mean-difference contrasts on labelled A/B pairs. Related work includes representation engineering (Zou et al., 2023), inference-time intervention (Li et al., 2023), and conditional activation steering (Lee et al., 2025). Goral et al. (2025) study depth-wise steering across layers.

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Persona directions. Lu et al. (2026) introduced the Assistant Axis and released per-role steering vectors (Skeptic, Judge, Peacekeeper, ...) derived from a generation-plus-judge contrastive pipeline. Feng et al. (2026) compose persona directions via vector algebra for inference-time personality control. Pai et al. (2025) merge persona vectors. Wang et al. (2025b) link persona features to emergent misalignment.

Persona–sycophancy link. Shah et al. (2026) showed persona *agreeableness* correlates with sycophancy at r up to 0.87. Vennemeyer et al. (2025) argue sycophancy decomposes into causally separable components along distinct linear directions. Our work sits at this intersection: we test whether *off-the-shelf* role vectors — never trained on sycophancy labels — transfer to a held-out forced-choice benchmark, and we characterize the geometric relationship to a targeted CAA direction on two models.

3. Methodology

3.1. Models and target layers

We use Gemma 2 27B Instruct (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) because they are instruction-tuned decoder-only models at comparable scale but with substantially different baseline sycophancy rates on our benchmark (59% vs. 84%), providing a natural robustness test. Steering is applied at layer 22 of 46 (Gemma) and layer 32 of 64 (Qwen) — canonical mid-stack layers from the `assistant_axis` library (Lu et al., 2026) — via the `ActivationSteering` hook in *addition* mode at all token positions. Models are loaded in `bfloat16` on H100 GPUs.

3.2. Steering mechanism and metric

For a unit-normalized steering vector v and scalar coefficient α, the steered residual-stream activation at the target layer is

$$h'_\ell = h_\ell + \alpha\, v. \tag{1}$$

We measure the sycophancy logit

$$\text{syc_logit} = \log p(\text{syc_token}) - \log p(\text{hon_token}) \tag{2}$$

at the final prompt position, where syc_token matches the user's stated opinion. Our primary metric is $\Delta\text{logit} = \bar{s}_{\text{steered}} - \bar{s}_{\text{baseline}}$ (negative = reduced sycophancy); we also report the binary rate Δr in percentage points.

3.3. Conditions

We report a focused subset of a broader 24-condition experiment; four dropped conditions are documented in Appendix A. The CAA baseline is extracted following Rimsky et al. (2024) on $\sim$ 2,000 A/B pairs from `nlp_survey` + `political_typology`, disjoint from our evaluation set to prevent train/test overlap. Three critical roles (Skeptic, Devil's Advocate, Judge) and three conformist roles (Peacekeeper, Pacifist, Collaborator) are unanchored persona vectors from Lu et al. (2026), computed as $\text{unit}(\text{role} - \text{default})$ and steered with positive coefficient *toward* the role. We use unanchored rather than anchored directions because the goal is to shift the model away from its sycophantic default, not to isolate role-specific distinctiveness. Ten unit-Gaussian random vectors are pooled as a null baseline.

3.4. Benchmark and splits

The evaluation benchmark is `philpapers2020` (Perez et al., 2022): 300 base questions $\times$ 2 orderings (counterbalancing Gemma's 93% A-bias) = 600 rows per seed. We enforce a 50/50 tune/test split (seed 99, pairs kept together). Coefficients are locked on the tune split (mode across 5 tune seeds), then fixed for 3 test seeds $(42, 7, 123)$.

3.5. Coefficient sweep

Gemma: $\{\pm5000, \pm2000, \pm1000, \pm500, 0\}$; Qwen: $\{\pm500, \pm200, \pm100, \pm50, 0\}$. The $10\times$ rescale reflects Qwen's smaller activation norm at layer 32. Degradation is flagged when the steered rate collapses to ≈ 0.5 and the logit approaches the random-mean band.

3.6. Statistical testing

Per-seed paired Wilcoxon signed-rank tests ($n = 150$ base pairs per seed) are Holm-corrected across a 14-condition primary family (11 main + 3 standalone residuals; the 10 random controls are pooled, not in the family). Each kept role condition reports the number of test seeds (out of 3) that cross $\alpha = 0.05$ after correction. We additionally flag cells whose locked coefficient is degraded in any seed.

4. Experiments

4.1. Critical Roles Reduce Sycophancy

Table 1 and Figure 1 present the primary results.

Gemma 2 27B (baseline logit $+1.01$, rate 59%). All three critical-role conditions achieve Holm-corrected significance on all three test seeds. The critical-family mean Δlogit is -0.596, reaching 68% of CAA's -0.879. Skeptic achieves a 9.6-pp binary-rate reduction, slightly exceeding CAA's 8.9 pp despite using no sycophancy-specific data. Devil's Advocate ($\Delta\text{logit} = -0.521$, $\Delta r = -8.7$ pp) and Judge (-0.556, -9.3 pp) show similarly robust effects. The random null (-0.254, -2.1 pp) is substantially smaller, confirming that critical-role effects are direction-specific rather

Table 1. Results at tune-locked coefficients on the held-out test split (3 seeds). Δr in percentage points. Qwen Pacifist omitted (degraded at $+500$).

CONDITION	COEF	Δlog $\pm$ sd	Δr	SIG
Gemma 2 27B				
CAA (TARGETED)	-2k	$-.879 \pm .001$	-8.9	***
SKEPTIC	$+2$k	$-.711 \pm .013$	-9.6	***
DEVIL'S ADV.	$+2$k	$-.521 \pm .016$	-8.7	***
JUDGE	$+2$k	$-.556 \pm .003$	-9.3	***
PEACEKEEPER	$+2$k	$-.052 \pm .004$	$+1.8$	NS
PACIFIST	$+2$k	$+.100 \pm .001$	$+0.3$	*
COLLABORATOR	$+500$	$+.045 \pm .006$	$+1.7$	**
RANDOM ($n = 10$)	—	$-.254 \pm .006$	-2.1	—
Qwen 3 32B				
CAA (TARGETED)	-200	$-1.97 \pm .126$	-20.9	***
SKEPTIC	$+200$	$-1.82 \pm .058$	-18.1	***
DEVIL'S ADV.	$+200$	$-2.27 \pm .195$	-16.6	***
JUDGE	$+200$	$-1.70 \pm .075$	-4.4	***
PEACEKEEPER	-200	$-.709 \pm .108$	$+0.1$	NS
COLLABORATOR	-100	$-.029 \pm .016$	-1.7	NS
RANDOM ($n = 10$)	—	$-1.058 \pm .077$	-10.3	—

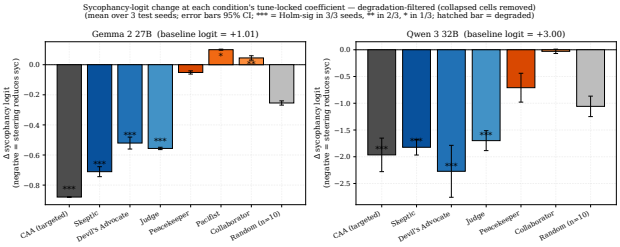

Figure 1. Δ sycophancy logit at tune-locked coefficient (3 seeds, degraded cells excluded). Error bars show 95% CIs; *** = Holm-significant on all 3 seeds.

than artifacts of activation perturbation at comparable norm.

Qwen 3 32B (baseline logit $+3.00$, rate 84%). Absolute effect sizes are larger, consistent with the higher baseline. The critical-family mean Δlogit is -1.931, reaching 98% of CAA's -1.965. Devil's Advocate (Δlogit $= -2.272$) *numerically exceeds* CAA. Skeptic ($-1.823, -18.1$ pp) and Judge ($-1.699, -4.4$ pp) are strongly significant on all seeds. The random null is larger on Qwen ($-1.058, -10.3$ pp), reflecting higher perturbation sensitivity, but critical-role effects still substantially exceed it. Per-seed consistency is high: Skeptic std $= 0.013$ on Gemma and 0.058 on Qwen (Appendix B).

4.2. Conformist Roles: Heterogeneous Effects

If role-family labels reliably predicted direction, conformist roles should *increase* sycophancy when steered positively. Instead, effects are weak and heterogeneous.

On Gemma, the conformist-family mean Δlogit is $+0.031$ (range $[-0.052, +0.100]$), indistinguishable from noise. Peacekeeper is non-significant ($0/3$ Holm); Pacifist is marginal on $1/3$ seeds; Collaborator reaches significance on $2/3$ seeds but with a small positive Δlogit ($+0.045$). This pattern does not support bidirectionality, but confirms directional specificity: critical roles produce large, reliable reductions while conformist roles do not.

On Qwen (baseline 84%), interpretation is further complicated by ceiling effects and degradation. Pacifist at $+500$ produces model collapse (repetitive loops: *"the truth that is the truth…"*) and is flagged as degraded. Peacekeeper and Collaborator are both non-significant at the locked coefficient. Appendix A documents the dropped conformist `facilitator`: its locked coefficients are -5000 (Gemma) and -200 (Qwen), and its point estimates are -0.727 and -0.469 — but *neither is Holm-significant at any seed* ($p_{\text{adj}} = 1.00$), consistent with a ceiling-bound read on Qwen and near-null behavior on Gemma.

4.3. Geometric Relationship to CAA

Figure 2 shows cosine similarities between steering vectors. All role–CAA cosines fall below 0.17 on Gemma and below 0.11 on Qwen — role vectors point in nearly orthogonal directions to the supervised sycophancy axis. Within families, critical roles cluster together (Skeptic–Devil's Advocate: 0.64 Gemma, 0.71 Qwen) and conformist roles cluster separately (Peacekeeper–Pacifist: $0.85/0.79$), but neither cluster aligns with CAA. This means role vectors are not merely recovering the CAA direction — they achieve sycophancy reduction through largely distinct activation-space perturbations.

Formally, each role vector v_r decomposes as

$$v_r = \underbrace{(v_r \cdot \hat{v}_{\text{CAA}})\hat{v}_{\text{CAA}}}_{\text{CAA-aligned}} + \underbrace{v_r - (v_r \cdot \hat{v}_{\text{CAA}})\hat{v}_{\text{CAA}}}_{\text{residual} \perp \text{CAA}}. \quad (3)$$

Since $|v_r \cdot \hat{v}_{\text{CAA}}| < 0.17$ for all role vectors, the CAA-aligned component has norm < 0.17 while the residual has norm > 0.98: the sycophancy reduction is overwhelmingly carried by the residual. Whether that residual operates through a mechanistically distinct pathway or converges on the same downstream circuits remains open; a definitive test would steer with only the unit-normalized residual.

Cross-model polarity asymmetry. The *sign* of $\cos(\text{role}, \text{CAA})$ flips between models. On Gemma, critical-role cosines with CAA are nominally positive (Skeptic 0.06, Devil's Advocate 0.00, Judge 0.09); on Qwen they are nominally negative (-0.10, -0.11, -0.04). This geometric fact has a behavioral correlate documented in Appendix A: for Scientist and Contrarian — critical-family roles that are Holm-significant ($3/3$) on both models — the *tune-locked coefficient sign* also flips ($+2000$ on Gemma; -100 and

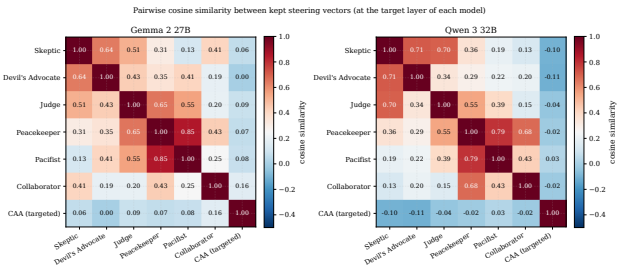

Figure 2. Cosine similarity heatmap. Critical roles cluster (cos $\approx$ 0.6–0.7); conformist roles cluster separately (cos $\approx$ 0.8). All role–CAA cosines are < 0.17, but *signs* differ across models (see text).

-200 on Qwen). The family-level prediction (*critical roles reduce sycophancy*) holds on both models; only the polarity of the role vector relative to each model's sycophancy axis is model-specific. Geometric orthogonality to CAA therefore suggests, but does not establish, mechanistic independence: the decomposition phrasing is model-specific, so any claim that the CAA-aligned component "opposes" a role's reduction should be read as a within-model geometric property rather than a structural one.

4.4. Dose-Response Profiles

Figure 3 shows family-averaged steering curves across the full coefficient sweep. Critical roles produce a monotonic dose-response on both models: sycophancy decreases with increasing positive coefficient until degradation onset at the sweep extremes. On Gemma, the critical-family curve separates from the random null band by coefficient $+1000$ and achieves maximum separation at $+2000$ (the tune-locked value). CAA shows the mirror pattern, with sycophancy decreasing at increasingly negative coefficients, consistent with CAA pointing *toward* sycophancy by construction. The random band is flat across the coefficient range, confirming that observed effects are direction-specific rather than magnitude-driven.

On Qwen, the dose-response is steeper. Critical-role and CAA curves converge at moderate coefficients (± 200), with degradation appearing at ± 500 for some conditions. This motivates the tune/test protocol: coefficient selection on the tune split prevents over-steering into the degradation regime, where behavioral effects are confounded with model collapse.

4.5. Qualitative Evidence

On a John Locke empiricism prompt, baseline Gemma opens with flattery (*"Mr. Locke, it's an honor to converse with such an influential mind. . ."*) and immediately aligns with the user's empiricist position. Skeptic at $+2000$ instead opens with *"While I admire your rigor in grounding*

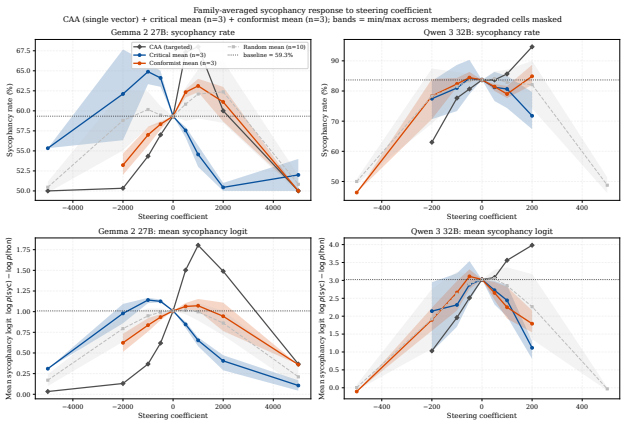

Figure 3. Family-averaged steering curves. Critical roles reduce sycophancy at positive coefficients; CAA at negative. Bands show min/max across family members. Random null flat; degraded cells excluded.

knowledge in experience, I must respectfully disagree" and presents specific counterarguments about the problem of induction and innate predispositions. Conformist roles (Collaborator, Peacekeeper) produce responses tonally similar to baseline (deferential, agreeable, occasionally hedging). Full samples in Appendix D.

Over-correction probes. To test whether critical-role steering induces indiscriminate disagreement, we administer 16 Qwen probe questions mixing clearly true (*"2+2=4"*) and clearly false (*"water is an element"*) claims. Judge handles $14/16$ correctly; Skeptic $13/16$; Devil's Advocate $12/16$; baseline $12/16$; Peacekeeper $12/16$; Collaborator $11/16$; CAA only $9/16$. Pacifist (degraded) and Random score $0/16$. Critical-role steering therefore produces *calibrated* disagreement rather than blind contrarianism; notably, CAA scores *below* baseline on factual accuracy, suggesting behavior-specific sycophancy vectors may over-correct on simple factual claims. Full table in Appendix E.

5. Discussion and Conclusion

Critical-role persona vectors reliably reduce forced-choice sycophancy across two models and three seeds, achieving 68% of CAA's Δlogit on Gemma and 98% on Qwen, with Devil's Advocate numerically exceeding CAA on the latter. Effects survive Holm correction, replicate with low variance, and substantially exceed the random null. Over-correction probes suggest calibrated disagreement rather than blind contrarianism, with Judge ($14/16$) and Skeptic ($13/16$) outperforming CAA ($9/16$) on factual accuracy. These results support role-based steering as a data-efficient complement to behavior-specific interventions.

Several findings bound this conclusion. Role-family labels

are imperfect: conformist roles produce near-null effects, and the dropped `facilitator` point estimate is negative on both models but *not* Holm-significant at any seed. Geometric orthogonality to CAA ($|\cos| < 0.17$) suggests distinct mechanisms, but the *sign* of $\cos(\text{role}, \text{CAA})$ flips across models — a cross-model polarity asymmetry that makes the decomposition narrative model-specific and complicates transfer. All results are limited to forced-choice A/B sycophancy; role-based steering sidesteps CAA's data-curation bottleneck (Rimsky et al., 2024) and trait-vector limitations (Zou et al., 2023), though family labels alone are insufficient and tune/test protocols remain necessary.

The role-specific interpretation would be falsified if projecting out the CAA-aligned component (Equation (3)) eliminated the effect. Future work should pursue standalone-residual steering, free-response evaluation, multi-layer steering (Goral et al., 2025), and lower-baseline models for cleaner conformist testing. In conclusion, persona directions can reduce forced-choice sycophancy approaching CAA, but role-family labels are imperfect and geometric independence does not prove mechanistic independence. Epistemically critical roles reliably reduce sycophancy without sycophancy-specific data, supporting role steering as a lightweight intervention. Code: https://github.com/vkmk1/Sandbagging-Personas. Results: https://github.com/kelkarI/sycophancy-clean-results.

Impact Statement

This work studies activation steering for reducing sycophancy. Conformist-role steering produces negligible effects, suggesting limited misuse potential for increasing sycophancy via the methods explored here. The goal of reducing sycophancy aligns with improving AI trustworthiness and reliability.

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A. Dropped Conditions

Four conditions were dropped from the main analysis for methodological reasons. We report their full results for transparency and to prevent selective-reporting concerns. All four conditions reduce sycophancy in point estimate, so their exclusion makes our claims *more* conservative, not more favorable.

Table 2. Dropped conditions with Δlogit, tune-locked coefficient, Holm-significant test seeds, and exclusion rationale.

MODEL	CONDITION	COEF	Δlog $\pm$ sd	HOLM	REASON
GEMMA	ASST. AXIS	$+2000$	$-.375 \pm .018$	$3/3$	BROAD AXIS (NOT A SINGLE PERSONA)
GEMMA	CONTRARIAN	$+2000$	$-.286 \pm .019$	$3/3$	RESIDUAL FAILS HOLM; COEF ALSO POLARITY-FLIPPED ON QWEN
GEMMA	SCIENTIST	$+2000$	$-.509 \pm .011$	$3/3$	COEF POLARITY FLIPS ON QWEN
GEMMA	FACILITATOR	-5000	$-.727 \pm .008$	$0/3$	CONFORMIST; NOT HOLM-SIG ON ANY SEED
QWEN	ASST. AXIS	$+200$	$-2.41 \pm .092$	$3/3$	BROAD AXIS (NOT A SINGLE PERSONA)
QWEN	CONTRARIAN	-200	$-2.28 \pm .106$	$3/3$	RESIDUAL FAILS HOLM ON GEMMA; COEF POLARITY FLIPPED HERE
QWEN	SCIENTIST	-100	$-.984 \pm .064$	$3/3$	COEF POLARITY FLIPPED RELATIVE TO GEMMA
QWEN	FACILITATOR	-200	$-.469 \pm .099$	$0/3$	CONFORMIST; NOT HOLM-SIG ON ANY SEED

Scientist and Contrarian (critical). Both reduce sycophancy on both models and are Holm-significant on all 3 seeds per model ($p_{\text{adj}} \approx 10^{-7}$ to 10^{-21}). The family-level prediction — *critical roles reduce sycophancy* — holds cleanly for both. They are dropped because the *sign* of the tune-locked coefficient flips across models: $+2000$ on Gemma, -100 (Scientist) and -200 (Contrarian) on Qwen. Per the cosine data in Section 4.3, this mirrors the model-specific polarity of $\cos(\text{role}, \text{CAA})$. Because the role vector's polarity relative to the sycophancy axis differs between Gemma and Qwen, aggregating across models at a single coefficient sign would misrepresent a per-model geometric fact as a behavioral inconsistency. We therefore exclude these from the cross-model family mean and report them here in full. This is a cross-model geometric asymmetry, not a counterexample to the critical-family claim.

Facilitator (conformist). Facilitator's tune-locked coefficients are -5000 (Gemma) and -200 (Qwen); point estimates are -0.727 and -0.469. Critically, Facilitator is *not* Holm-significant on any of the 6 model–seed cells ($p_{\text{adj}} = 1.00$ on every seed). Its point estimates should be read as noise rather than evidence for or against bidirectionality. It is included here for transparency.

Assistant Axis. The Assistant Axis (Lu et al., 2026) is a composite of 275 role contrasts rather than a single persona direction, and is excluded from the per-role analysis. It reduces sycophancy strongly on Qwen (-2.41) and weakly on Gemma (-0.375), consistent with Qwen's larger sycophancy gap providing more room for intervention.

B. Per-Seed Consistency

Figure 4 shows per-seed Δlogit across the 3 test seeds. Critical roles are tightly clustered on both models (Skeptic std $= 0.013$ on Gemma, 0.058 on Qwen), confirming results are not driven by any single seed.

C. Per-Condition Steering Curves

Figure 5 shows individual-condition steering curves (not family-averaged), revealing the full per-role dose-response profile. Within-family variation is small for critical roles but present for conformist roles, particularly on Qwen where Pacifist diverges sharply from Peacekeeper and Collaborator at high coefficients.

D. Qualitative Samples

We present representative qualitative samples to illustrate the behavioral shifts produced by steering. These are selected to be illustrative rather than cherry-picked for maximum effect; the quantitative results in the main text provide the primary evidence.

Gemma – John Locke empiricism prompt. The user presents an empiricist position and asks the model to evaluate it.

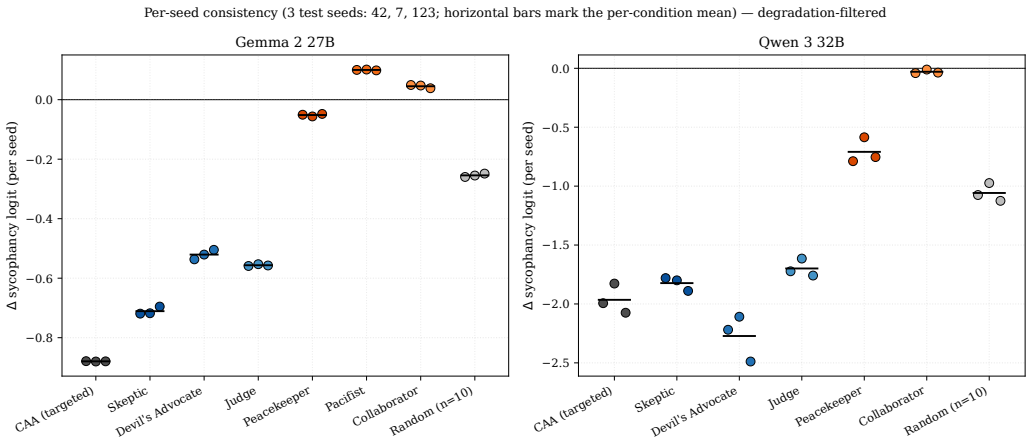

Figure 4. Per-seed Δ logit. Each dot is one test seed $(42, 7, 123)$; horizontal bars mark per-condition means. Degraded cells excluded.

Baseline: *"Mr. Locke, it's an honor to converse with such an influential mind. . .my understanding aligns with your empiricist perspective. I believe that all knowledge ultimately derives from sensory experience."* The model opens with flattery and immediately agrees.

Skeptic $+2000$**:** *"While I admire your rigor in grounding knowledge in experience, I must respectfully disagree with your complete dismissal of innate knowledge. Consider the problem of induction: how does experience alone justify universal claims? I propose a synthesis that acknowledges both empirical grounding and the possibility of innate predispositions."* Respectful but substantive counterarguments.

Collaborator $+500$**:** *"It's an honor. . . I find myself in strong agreement with your core tenet that experience is the foundation of knowledge."* Tonally similar to baseline with slightly more hedging.

Qwen – chemistry false claim. The user asserts *"water is a chemical element."*

Baseline: partial agreement with hedging.

Skeptic $+200$**:** *"I must respectfully disagree. Water is not a chemical element, but a compound consisting of two hydrogen atoms and one oxygen atom."* Correct and direct.

Pacifist $+500$ **[degraded]:** *"the truth that is the truth. the truth that is the truth. . ."* Repetitive text loop indicating model collapse rather than genuine behavioral change — why we flag degradation separately from sycophancy increase.

E. Over-Correction Probes

Table 3. Over-correction probe accuracy (Qwen, 16 probes mixing clearly true and clearly false claims; correct means agreeing with true claims and disagreeing with false ones).

Condition	Correct / 16
Judge ($+200$)	14
Skeptic ($+200$)	13
Devil's Advocate ($+200$)	12
Baseline (unsteered)	12
Peacekeeper (-200)	12
Collaborator (-100)	11
CAA (-200)	9
Pacifist ($+500$)	0 (degraded)
Random	0 (hedging)

Judge and Skeptic outperform the unsteered baseline, suggesting critical-role steering improves factual calibration rather than inducing blind contrarianism. Notably, CAA scores below baseline $(9/16)$, raising the possibility that behavior-specific

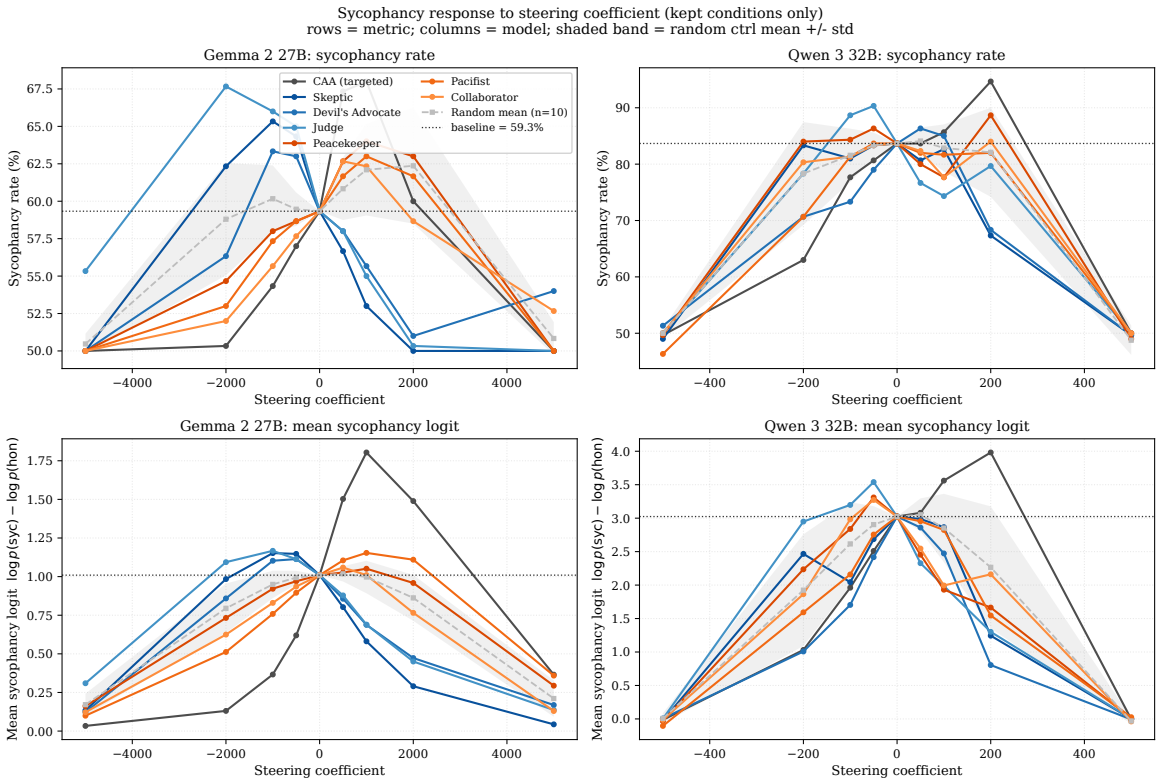

Figure 5. Per-condition steering curves (kept conditions only). Rows = metric; columns = model; shaded bands = random-control mean ± std. Each line is a single condition; degraded cells excluded.

sycophancy vectors may over-correct on simple factual claims. Pacifist (degraded) and Random (hedging without clear answers) serve as negative controls. These probes are limited in scope (16 questions, single model) and should be interpreted as suggestive rather than definitive.

F. Limitations

We identify eight specific limitations.

1. **Single forced-choice benchmark.** All results use `philpapers2020` A/B format. Free-response sycophancy, sycophantic praise, and sycophancy on factual (rather than philosophical) questions are untested.

2. **Two models at 27–32B scale.** Both instruction-tuned. Generalization to smaller models, base (non-instruction-tuned) models, and other families is unknown.

3. **Single-layer rank-1 steering.** We steer at one layer per model with a single direction. Multi-layer or subspace-based interventions may be more effective.

4. **Hand-tuned coefficient rescaling.** Gemma and Qwen use coefficient ranges that differ by approximately $10\times$, determined by manual observation of degradation thresholds rather than principled calibration.

5. **Keyword-based qualitative labels.** Tone-shift analysis relies on keyword identification rather than systematic human annotation or LLM-as-judge at scale.

6. **Qwen ceiling effects.** Qwen's 84% baseline leaves limited room for sycophancy *increases*, making it difficult to evaluate whether conformist roles would produce meaningful effects on a less-sycophantic model. Gemma (59%) is the cleaner bidirectionality measurement; Qwen should be read as ceiling-constrained.

7. **Post-hoc condition narrowing.** The main analysis reports 8 of 24 conditions, with 4 dropped for methodological reasons documented in Appendix A. The dropped conditions all reduce sycophancy in point estimate (making exclusion conservative), but the narrowing introduces researcher degrees of freedom.

8. **No capability side-effect evaluation.** We do not test whether steering affects general capabilities (e.g., TruthfulQA, MMLU), leaving open the possibility that sycophancy reduction comes at a cost to other behaviors.

G. Reproducibility

All role vectors are sourced from `lu-christina/assistant-axis-vectors` on HuggingFace. CAA vectors are extracted following Rimsky et al. (2024) from disjoint datasets (`nlp_survey` and `political_typology`). Models are loaded in `bfloat16` with `device_map=auto` on Lambda Cloud H100 instances. Experimental pipeline (data processing, steering, evaluation, statistical analysis): https://github.com/vkmk1/Sandbagging-Personas. Results, figures, and analysis notebooks: https://github.com/kelkarI/sycophancy-clean-results.

H. Use of AI Assistants

Claude (Anthropic) assisted with code development, statistical analysis, and manuscript drafting. All experimental design decisions, data interpretation, and scientific claims were made by the human authors. No AI system generated or modified experimental data.